

Docon

Prepared by

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Abstract

Docon is an digital healthcare platform designed to simplify patient management, telemedicine consultations, and medical record-keeping for doctors, and hospitals. This system integrates online appointment scheduling, electronic health records (EHR), e-prescriptions, and remote consultations into a seamless and secure interface. By leveraging advanced technology, Docon enhances efficiency in healthcare delivery, improves patient engagement, and streamlines doctor-patient interactions.

This project aims to bridge the gap between patients and healthcare providers, enabling accessible, efficient, and data-driven medical care.

Objectives of the Project

The primary objective of the Docon digital healthcare platform is to streamline and enhance the delivery of healthcare services by integrating advanced digital tools for patient management, telemedicine, electronic health records (EHR), and e-prescriptions. The project aims to bridge the gap between patients and healthcare providers by enabling a secure, accessible, and data-driven environment that fosters efficient medical care, improves patient engagement, and supports informed clinical decision-making.

Project Goals

- Enhanced healthcare accessibility through telemedicine.
- Efficient patient management with secure digital health records.
- Reduced waiting times and improved doctor-patient interactions.

Core Functionalities

- Online Appointment Scheduling & Queue Management

Appointment booking system with time slot optimization.

Real-time updates and notifications for upcoming consultations.

- Electronic Health Records (EHR) & e-Prescriptions

Secure storage of patient medical history and prescriptions.

Insights for diagnosis and treatment recommendations.

Seamless sharing of medical data with authorized doctors.

- Health Insights & Reminders

Personalized health recommendations based on medical history.

Medication reminders and follow-up alerts.

Predictive analytics for chronic disease management.

- Billing

Automated payment processing.

Integration with third-party payment gateways for seamless transactions.

- Telemedicine & Remote Consultations

Chat-based virtual doctor consultations.

Symptom checker for Preliminary assessments.

scope

- Doctors

Manage patient records and medical history digitally.

Conduct remote consultations via chat.

Generate and share digital prescriptions securely to patient.

- Patients

Book online or in-clinic doctor appointments.

Access personal medical records and prescriptions.

- Clinics & Hospitals

Automate patient data management and billing.

Optimize appointment scheduling and queue management.

Technologies

- Technology:- MERN
- Fronted:-React js, vite
- Backend:-Node js , Express js
- Database:-MongoDB
- Server:-Express server

Workflow

Patients register and book consultations through the platform.

Doctors review patient history and conduct in-person or virtual consultations.

Digital prescriptions and medical records are securely stored and shared.

Patients receive follow-up reminders and health insights for better care management.

Implementation

System Architecture & Design

- Backend: Node.js | Frontend: React
- Responsive UI for desktop

User & Data Management

- Role-based access for doctors, admin, patients
- JWT for secure login

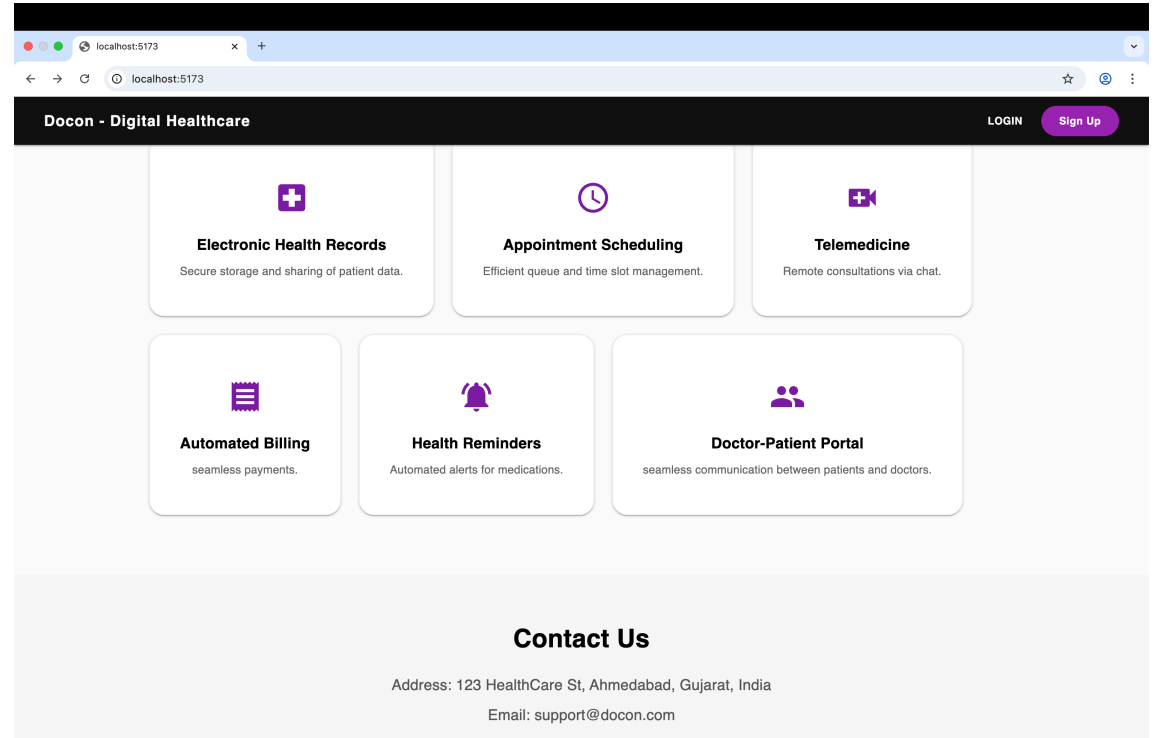
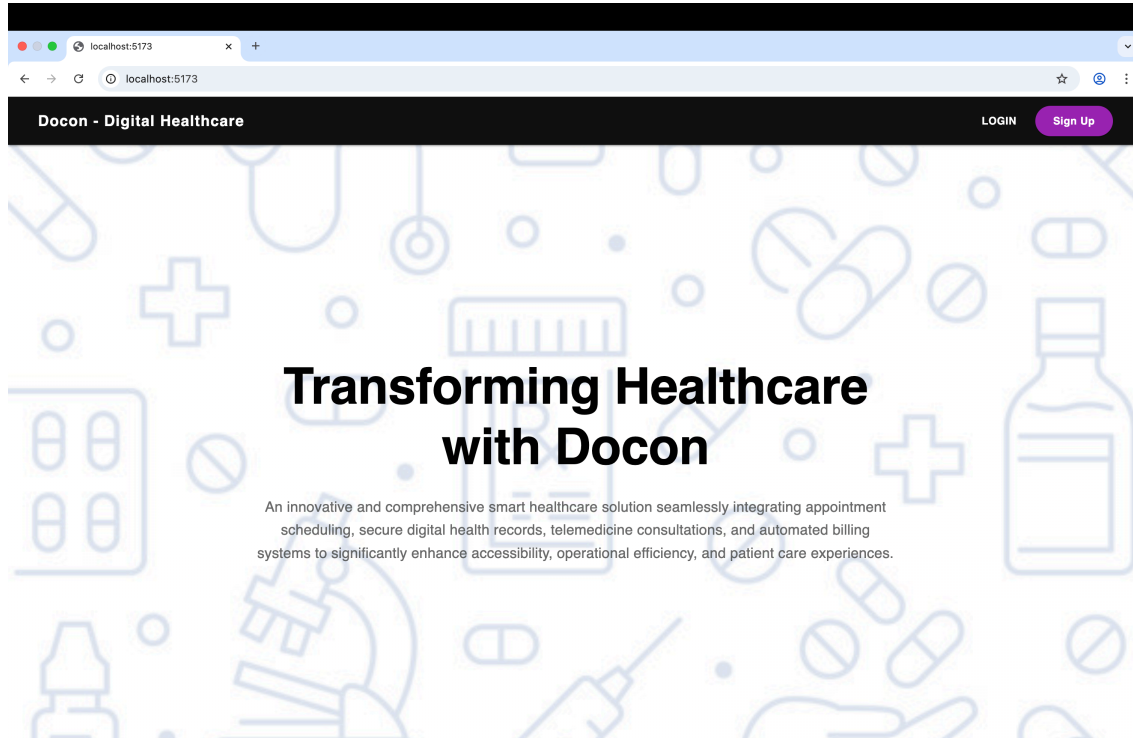
Key Features

- Online appointment scheduling with real-time
- Telemedicine via secure chat
- Digital prescriptions
- EHR for centralized medical records

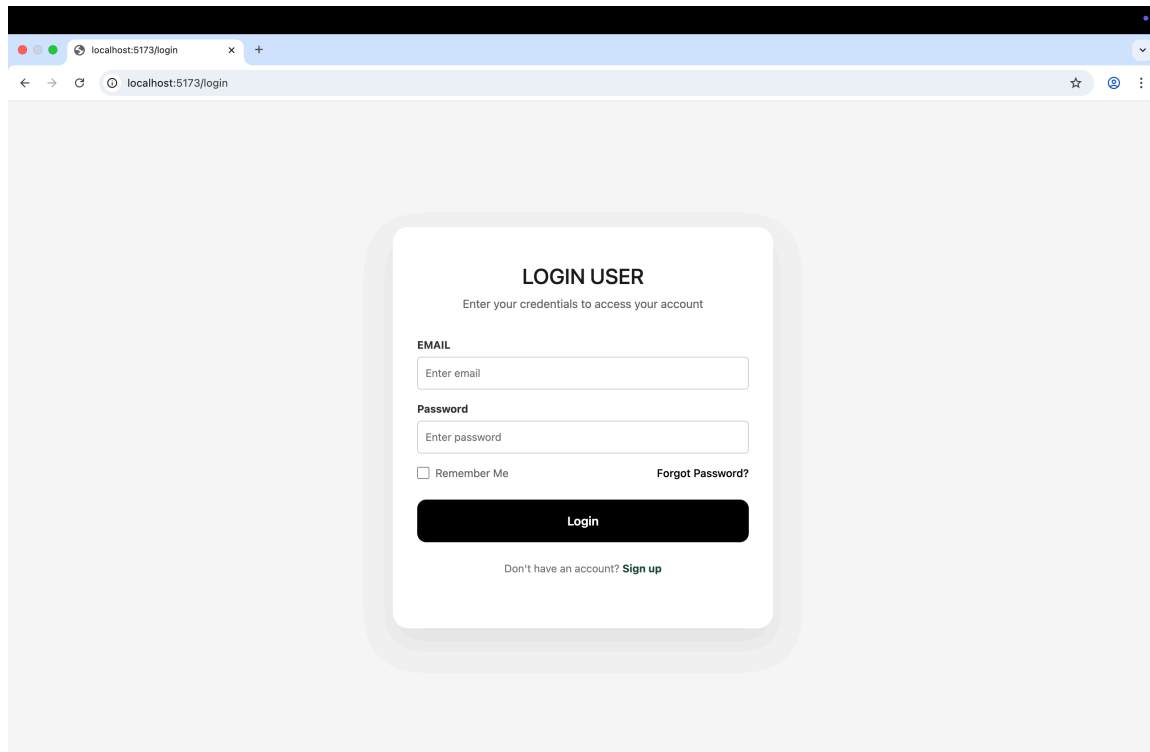
Interoperability & Analytics

- Dashboard for doctors, admin, patients

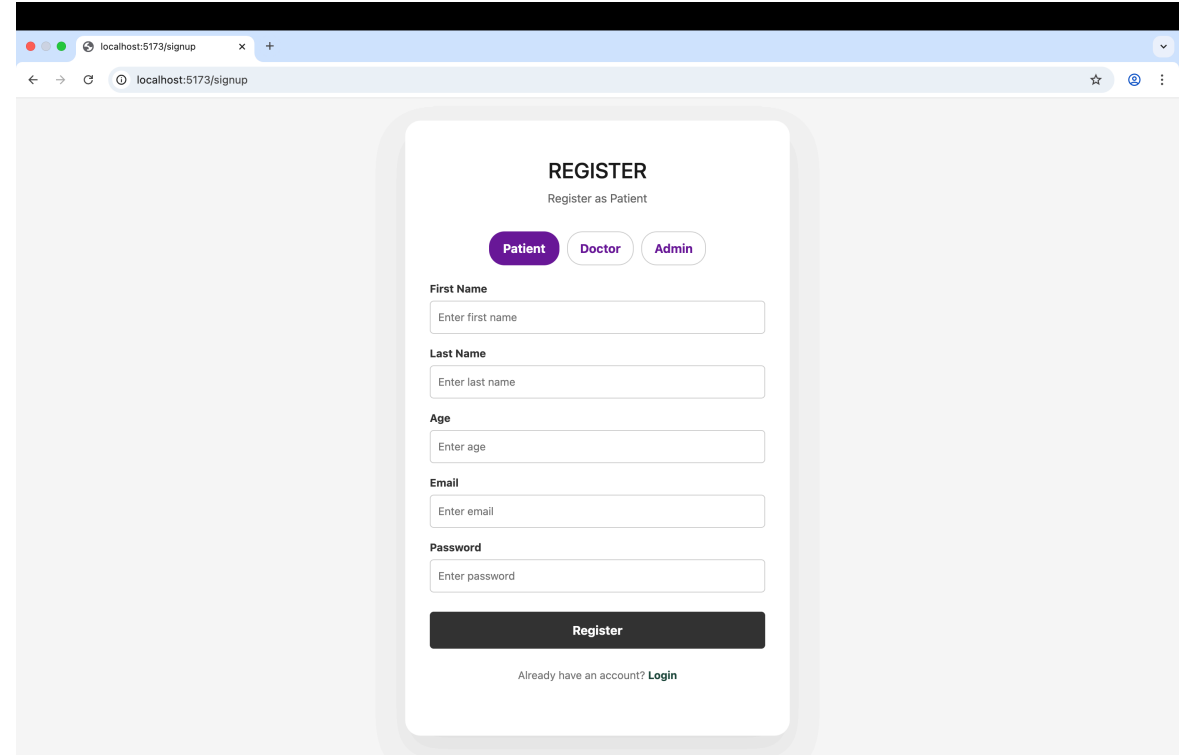
Result screen shots



Login & register for doctor, patient, admin



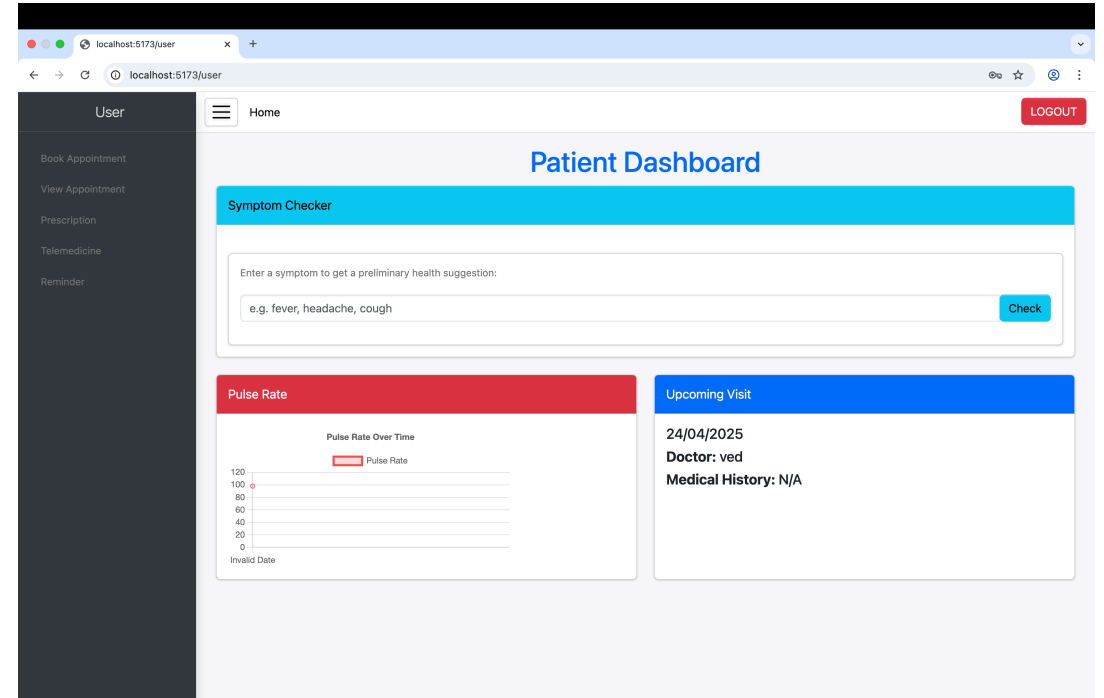
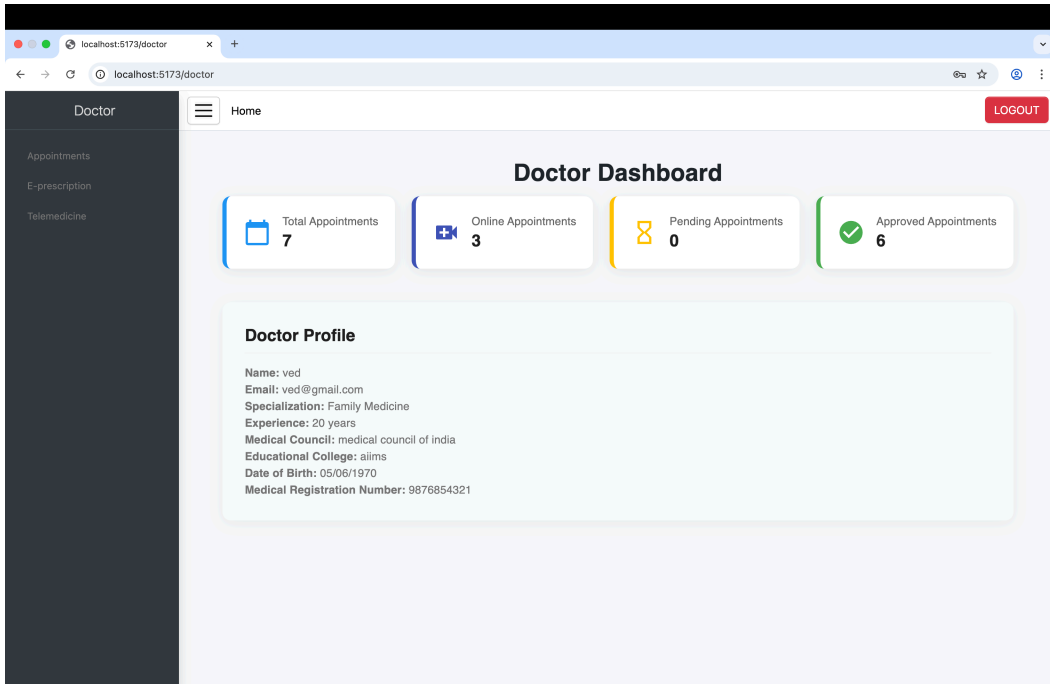
The screenshot shows a web browser window with the address bar displaying 'localhost:5173/login'. The page features a central white card with a light gray shadow. At the top of the card is the heading 'LOGIN USER' followed by the instruction 'Enter your credentials to access your account'. Below this, there are two input fields: 'EMAIL' with the placeholder 'Enter email' and 'Password' with the placeholder 'Enter password'. A checkbox labeled 'Remember Me' is positioned to the left of a link 'Forgot Password?'. A large black button with the text 'Login' in white is centered below the inputs. At the bottom of the card, a link 'Don't have an account? Sign up' is displayed.



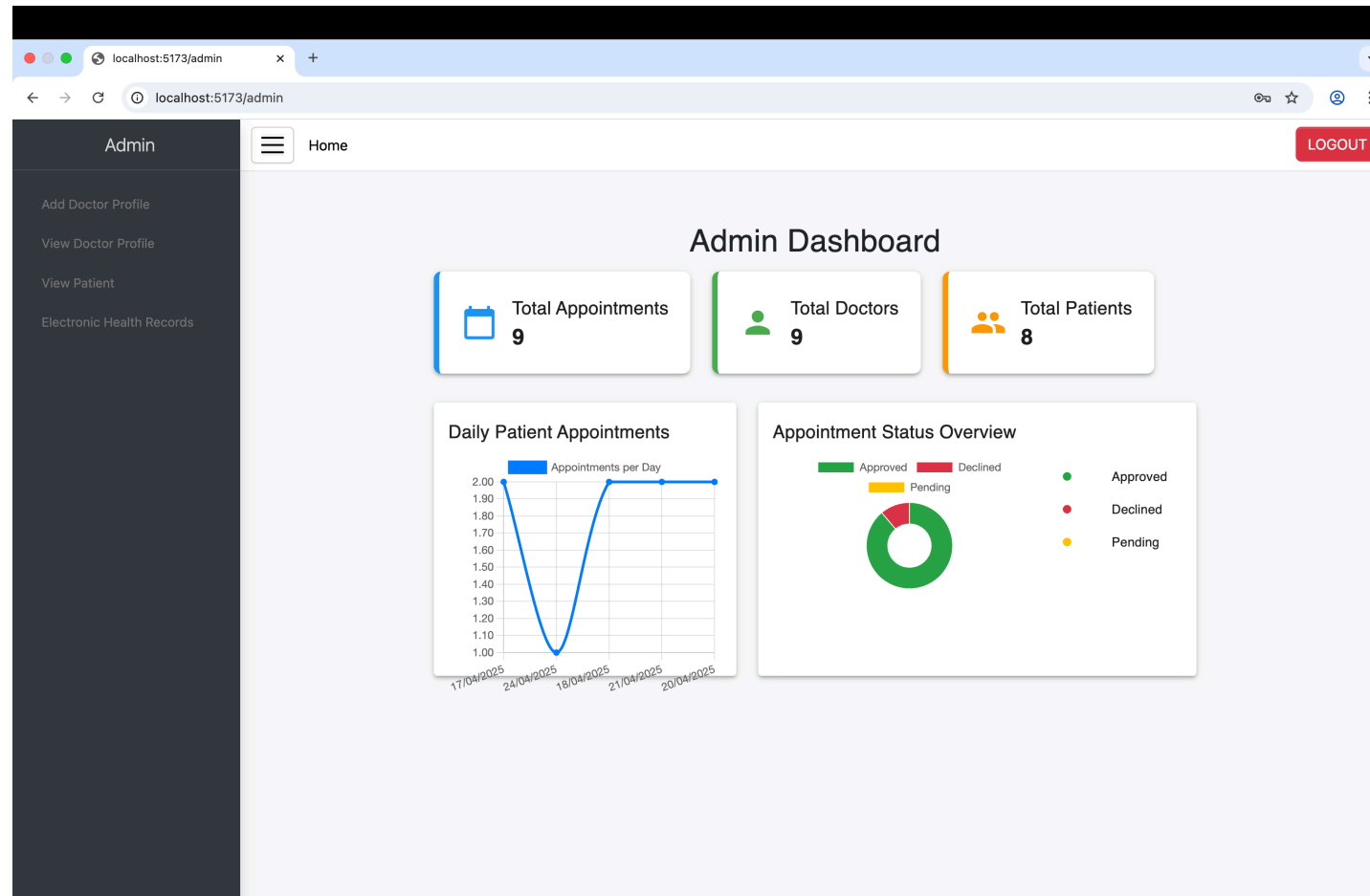
The screenshot shows a web browser window with the address bar displaying 'localhost:5173/signup'. The page features a central white card with a light gray shadow. At the top of the card is the heading 'REGISTER' followed by the instruction 'Register as Patient'. Below this, there are three buttons: 'Patient' (highlighted in purple), 'Doctor', and 'Admin'. The form includes several input fields: 'First Name' (placeholder 'Enter first name'), 'Last Name' (placeholder 'Enter last name'), 'Age' (placeholder 'Enter age'), 'Email' (placeholder 'Enter email'), and 'Password' (placeholder 'Enter password'). A large black button with the text 'Register' in white is centered below the inputs. At the bottom of the card, a link 'Already have an account? Login' is displayed.

Allows users (doctors, admin and patients) to securely access the platform by logging in or creating a new account with essential credentials.

Dashboard



Provides doctors with tools for appointments details , and doctor profile, while patients can view symptom checker, pulse rate.



The admin dashboard displays key metrics: Total Appointments, Total Doctors, Total Patients, and Daily Patient Appointments for quick insights. An Appointment Status Overview Chart visualizes statuses like *Pending*, *approved*, and *Cancelled* to monitor performance at a glance.

Booking appointment

The screenshot shows a web browser at localhost:5173/user/addappointment. The page has a dark sidebar with links: User, Book Appointment, View Appointment, Prescription, Telemedicine, and Reminder. The main content area has a 'Home' link and a 'LOGOUT' button. The 'Appointment' form is centered and contains the following fields:

- Patient Name:
- Patient Age:
- Select Doctor:
- Appointment Date: with a calendar icon
- Select Time Slot:
- Appointment Type:
- Reference:
- Submit:

The screenshot shows a web browser at localhost:5173/user/viewappointment. The page has a dark sidebar with links: User, Book Appointment, View Appointment, Prescription, Telemedicine, and Reminder. The main content area has a 'Home' link and a 'LOGOUT' button. The 'My Appointments' table is displayed with the following data:

Doctor Name	Specialization	Date	Time	Apointment Type	Status
ved	Family Medicine	17/04/2025	10:00 AM	Offline	Approved
ved	Family Medicine	24/04/2025	04:00 PM	Online	Approved

At the bottom right of the table, there is a pagination control: Rows per page: 100, 1-2 of 2, and navigation arrows.

The Book Appointment page allows users to select a doctor, date, and time to schedule a medical visit. The View Appointment Status page shows upcoming and past appointments with real-time status updates like Pending, approved, or declined.

Doctor appointments and e-prescription

The screenshot shows the 'E-Prescription Portal' interface. On the left is a dark sidebar with 'Doctor' at the top and three menu items: 'Appointments', 'E-prescription', and 'Telemedicine'. The main content area has a header 'E-Prescription Portal'. Below it is a 'Select Appointment' section with an 'Appointment Date' field containing '20/04/2025' and a 'Select Patient' dropdown menu showing 'rathin'. Below this is a 'Write Prescription for rathin' section with three text input fields for 'Symptoms', 'Diagnosis', and 'Pulse Rate'. Underneath is a 'Medications' section with a table-like structure for adding medications, including fields for 'Name', 'Dosage', 'Quantity', and 'Instructions', and a '+ Add Medication' button. At the bottom of this section are two more text input fields for 'dd/mm/yyyy' and 'Medical History Notes', and a 'Billing Amount (₹)' field. A blue 'Submit Prescription' button is at the very bottom.

Doctor

Appointments

E-prescription

Telemedicine

E-Prescription Portal

Select Appointment

Appointment Date

20/04/2025

Select Patient

rathin

Write Prescription for rathin

Symptoms

Diagnosis

Pulse Rate

Medications

Name	Dosage	Quantity	Instructions
+ Add Medication			

dd/mm/yyyy

Medical History Notes

Billing Amount (₹)

Submit Prescription

The screenshot shows the 'My Appointments' interface. On the left is a dark sidebar with 'Doctor' at the top and three menu items: 'Appointments', 'E-prescription', and 'Telemedicine'. The main content area has a header 'My Appointments' with a 'Home' link and a 'LOGOUT' button. Below the header are four filter tabs: 'Pending', 'Approved', 'Declined', and 'All'. Below the filters is a table with columns: 'Patient Name', 'Age', 'Appointment Date', 'Appointment Time', 'Appointment Type', and 'Status'. The table contains eight rows of appointment data.

Doctor

Appointments

E-prescription

Telemedicine

My Appointments

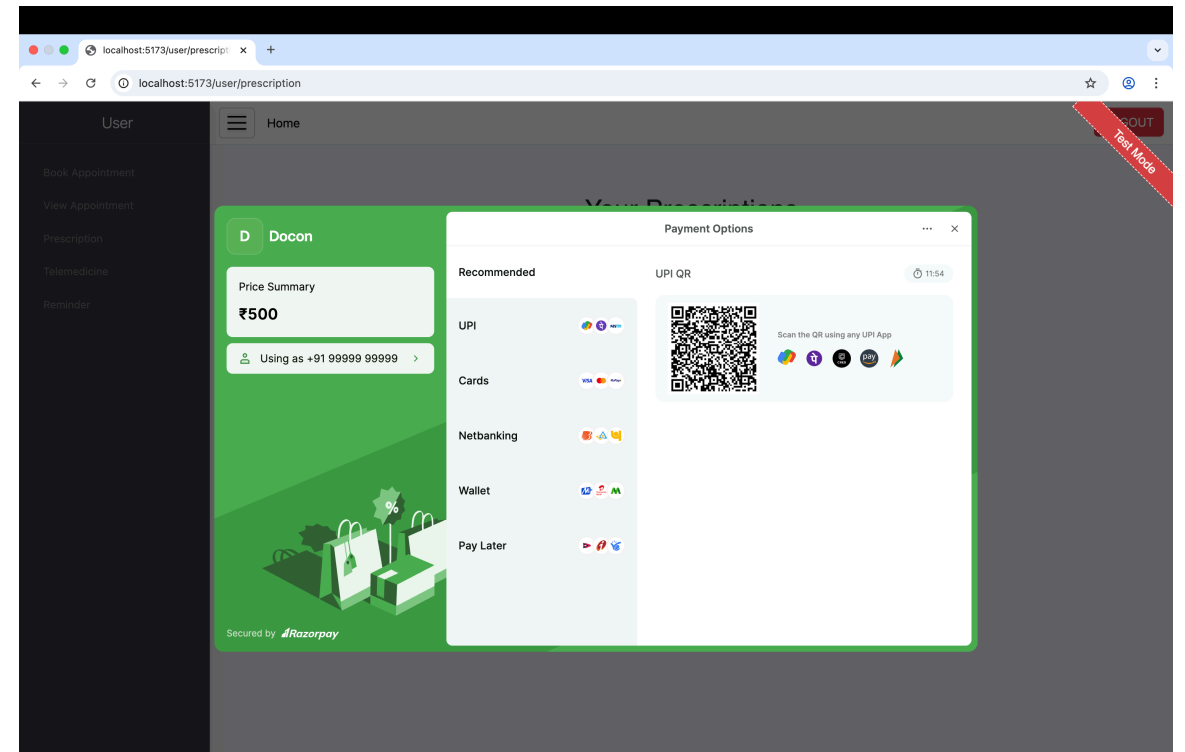
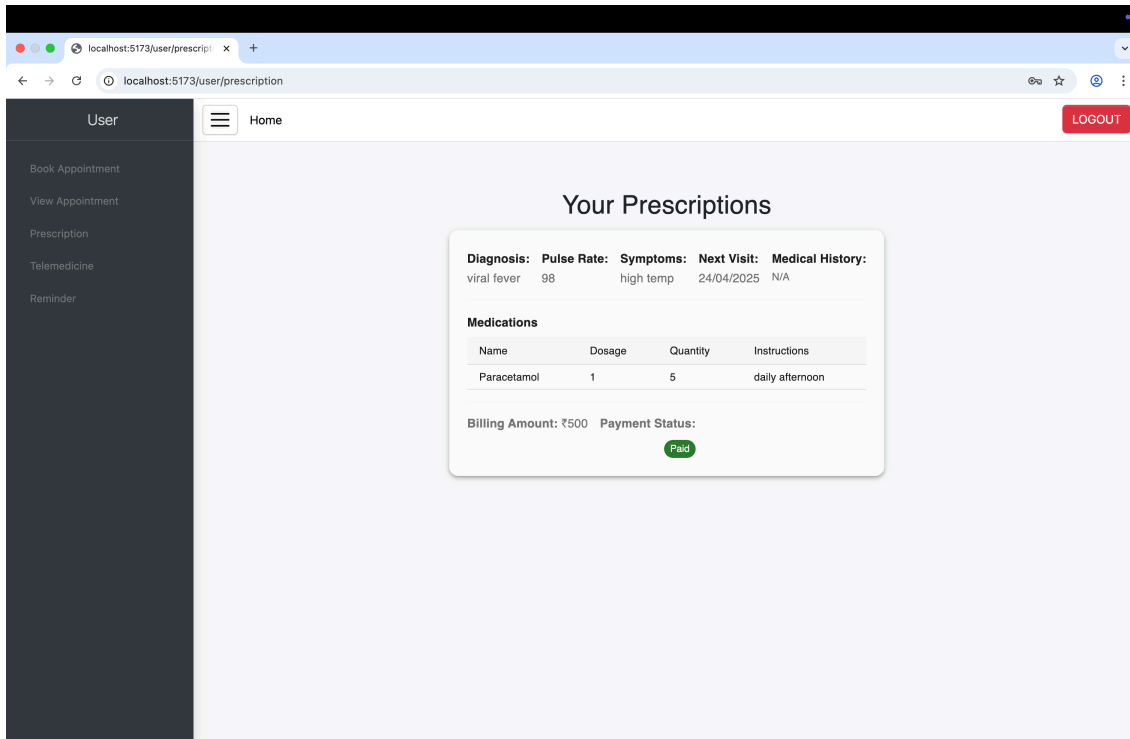
Home

LOGOUT

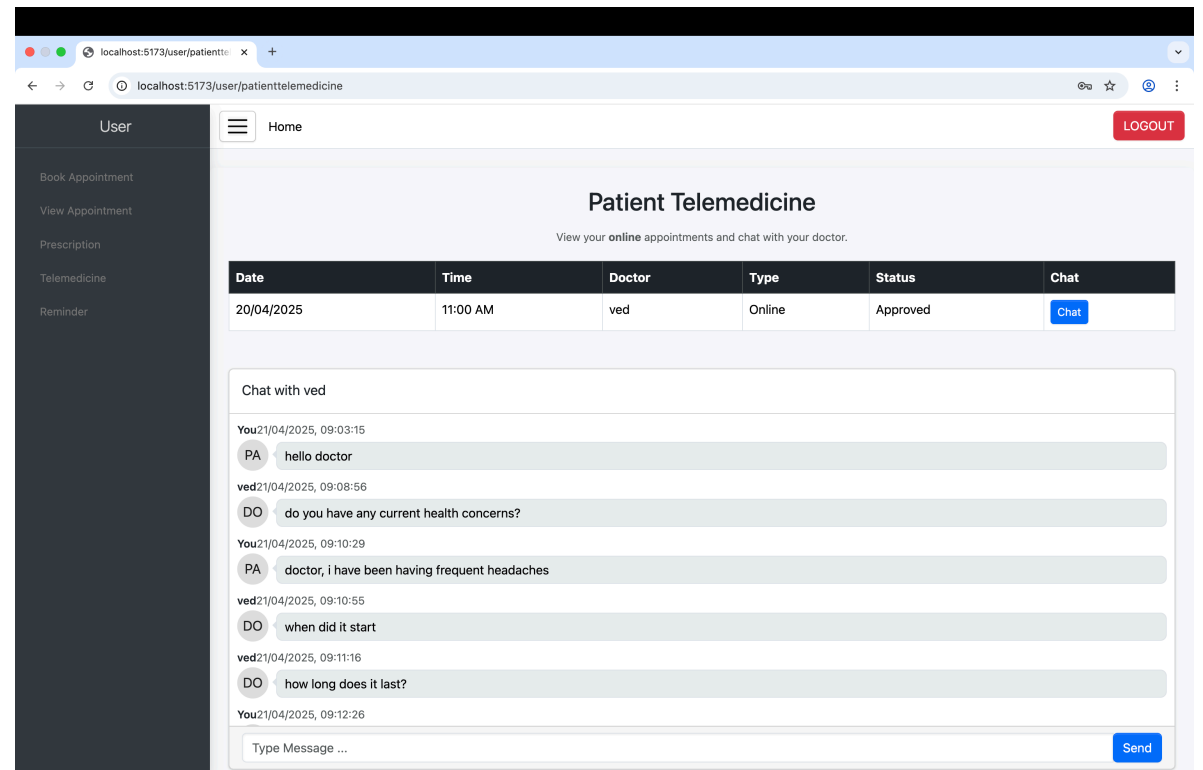
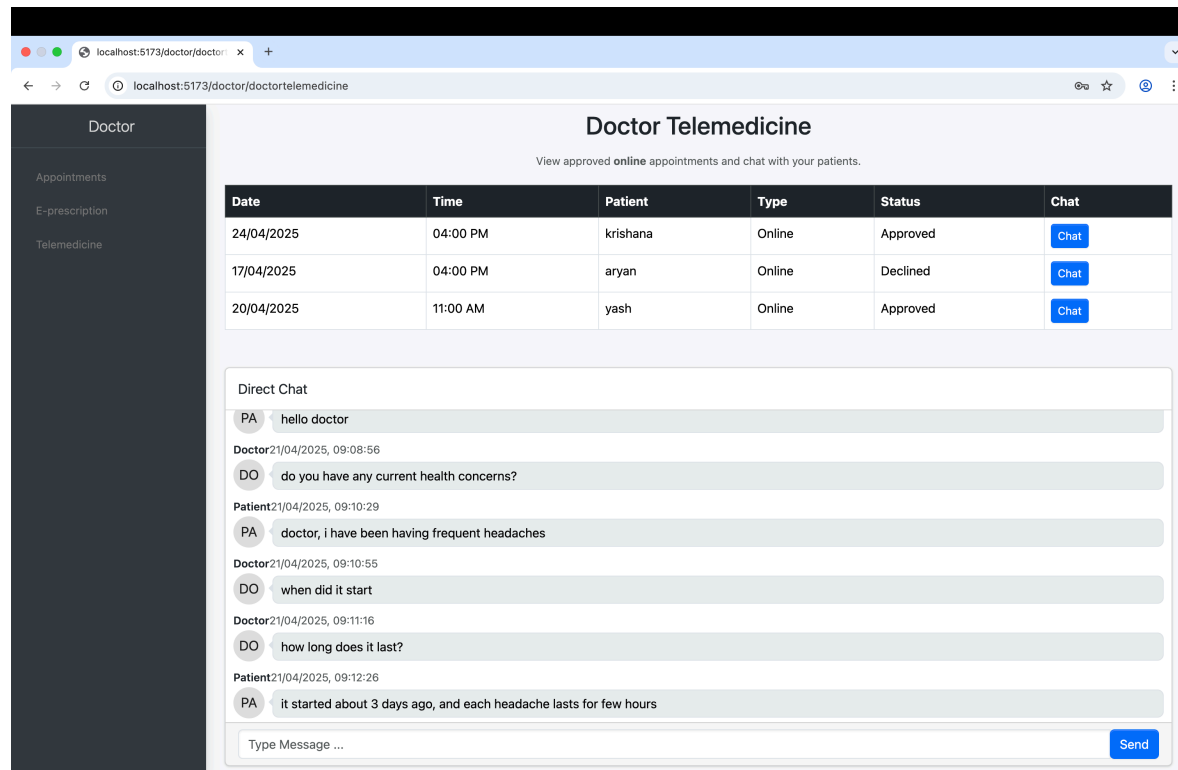
Pending Approved Declined All

Patient Name	Age	Appointment Date	Appointment Time	Appointment Type	Status
krishana	23	17/04/2025	10:00 AM	Offline	Approved
krishana	23	24/04/2025	04:00 PM	Online	Approved
aryan	45	17/04/2025	04:00 PM	Online	Declined
aryan	46	18/04/2025	05:00 PM	Offline	Approved
rathin	13	20/04/2025	04:00 PM	Offline	Approved
yash	19	20/04/2025	11:00 AM	Online	Approved
dharmik	19	21/04/2025	04:00 PM	Offline	Approved

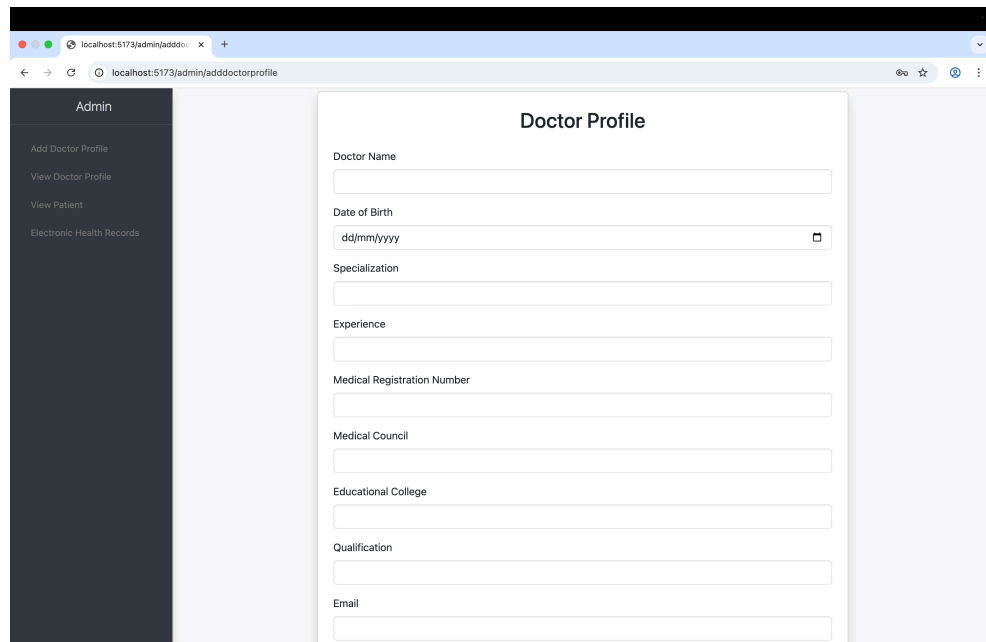
User prescription and payment gateway



Chat base Telemedicine



Doctor profile add and view



The screenshot shows the 'Add Doctor Profile' form in the Admin panel. The form is titled 'Doctor Profile' and contains several input fields for doctor information. The left sidebar shows the Admin menu with options: Add Doctor Profile, View Doctor Profile, View Patient, and Electronic Health Records.

Admin

- Add Doctor Profile
- View Doctor Profile
- View Patient
- Electronic Health Records

Doctor Profile

Doctor Name

Date of Birth

Specialization

Experience

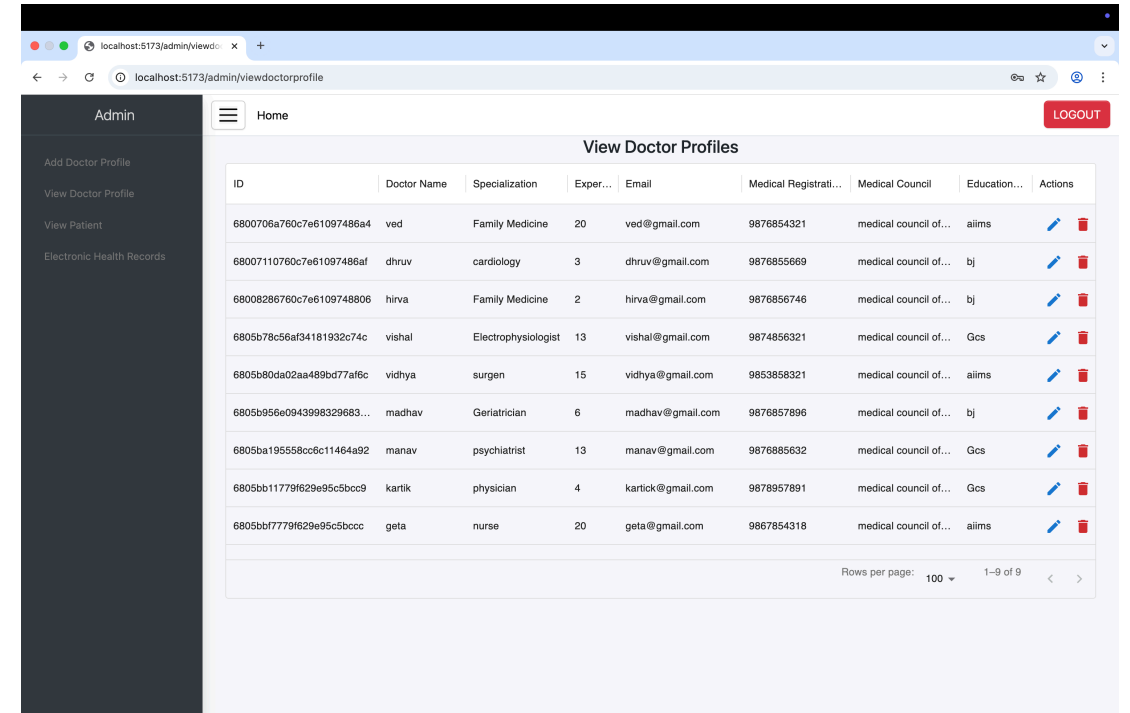
Medical Registration Number

Medical Council

Educational College

Qualification

Email



The screenshot shows the 'View Doctor Profiles' table in the Admin panel. The table displays a list of doctors with their profiles. The left sidebar shows the Admin menu with options: Add Doctor Profile, View Doctor Profile, View Patient, and Electronic Health Records. The top right corner has a 'LOGOUT' button.

Admin

- Add Doctor Profile
- View Doctor Profile
- View Patient
- Electronic Health Records

View Doctor Profiles

ID	Doctor Name	Specialization	Exper...	Email	Medical Registrati...	Medical Council	Education...	Actions
6800706a760c7e61097486a4	ved	Family Medicine	20	ved@gmail.com	9876854321	medical council of...	alims	Edit Delete
68007110760c7e61097486af	dhruv	cardiology	3	dhruv@gmail.com	9876855669	medical council of...	bj	Edit Delete
68008286760c7e6109748806	hirva	Family Medicine	2	hirva@gmail.com	9876856746	medical council of...	bj	Edit Delete
6805b78c56af34181932c74c	vishal	Electrophysiologist	13	vishal@gmail.com	9874856321	medical council of...	Gcs	Edit Delete
6805b80da02aa489bd77af6c	vidhya	surgen	15	vidhya@gmail.com	9853858321	medical council of...	alims	Edit Delete
6805b956e0943998329683...	madhav	Geriatrician	6	madhav@gmail.com	9876857896	medical council of...	bj	Edit Delete
6805ba19558bcc5c11464a92	manav	psychiatrist	13	manav@gmail.com	9876885632	medical council of...	Gcs	Edit Delete
6805bb11779f629e95c5bcc9	kartik	physician	4	kartick@gmail.com	9878957891	medical council of...	Gcs	Edit Delete
6805bbf7779f629e95c5bccc	geta	nurse	20	geta@gmail.com	9867854318	medical council of...	alims	Edit Delete

Rows per page: 100 1-9 of 9

Update doctor Profile

localhost:5173/admin/update-doctor/6800706a760c7e61097486a4

Admin

- Add Doctor Profile
- View Doctor Profile
- View Patient
- Electronic Health Records

Home

LOGOUT

Update Doctor Profile

Doctor Name *
ved

Specialization *
Family Medicine

Experience *
20

Email *
ved@gmail.com

Medical Registration Number *
9876854321

Medical Council *
medical council of india

Educational College *
aiims

date of birth
05/06/1970

UPDATE

Electronic health records

The screenshot displays a web application for managing Electronic Health Records (EHR). The interface is divided into a left sidebar for administration and a main content area for patient records.

Admin Sidebar:

- Admin
- Add Doctor Profile
- View Doctor Profile
- View Patient
- Electronic Health Records

Electronic Health Records - All Patients

Search by Patient or Doctor Name

Patient: krishana **Doctor: ved** **Date Issued: 17/04/2025** **Diagnosis: viral fever**
Email: krishana@gmail.com **Next Visit: 24/04/2025** **Symptoms: high temp**
Pulse Rate: 98

Medications

Name	Dosage	Quantity	Instructions
Paracetamol	1	5	daily afternoon

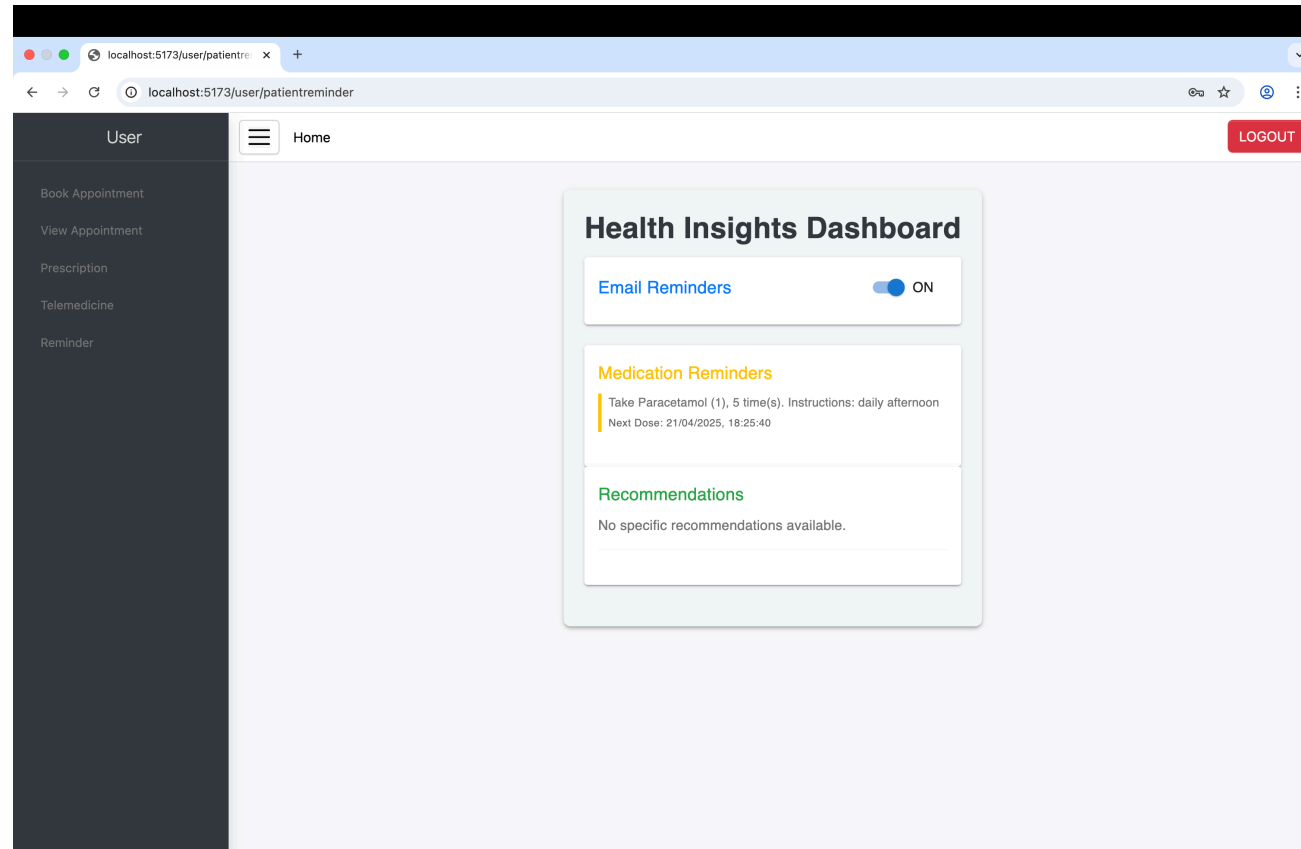
Patient: ganpat **Doctor: dhruv** **Date Issued: 17/04/2025** **Diagnosis: hypertension**
Email: ganpat@gmail.com **Next Visit: 18/05/2025** **Symptoms: blood pressure consistently high**
Pulse Rate: 128

Medications

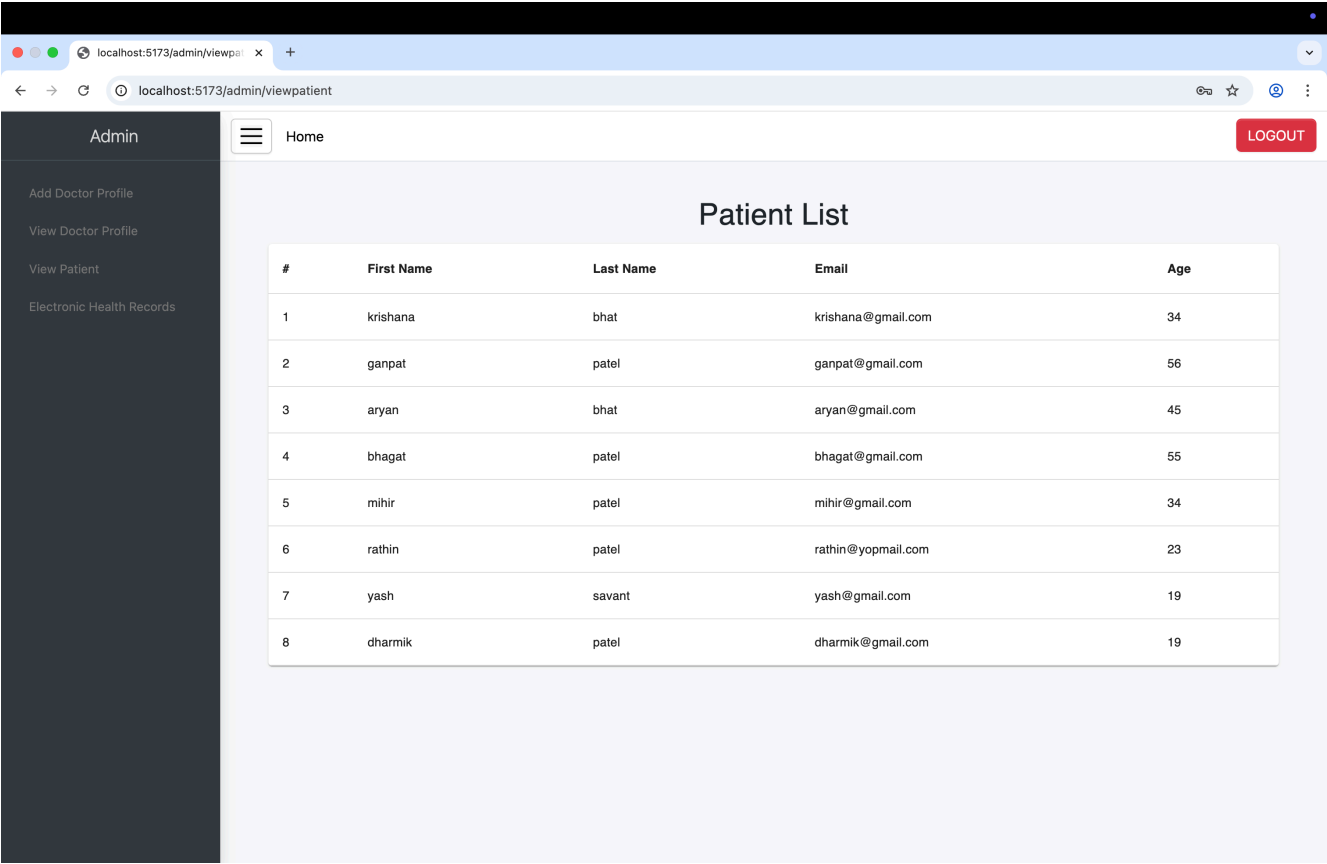
Name	Dosage	Quantity	Instructions
diuretics	1	30	daily morning
ACE inhibitors	1	15	daily morning

Patient: bhagat **Doctor: dhruv** **Date Issued: 21/04/2025** **Diagnosis: CAD**
Email: bhagat@gmail.com **Next Visit: Not Scheduled** **Symptoms: chest pain**

Health reminder



Register Patient list



The screenshot shows a web browser window with the address bar displaying 'localhost:5173/admin/viewpatient'. The page has a dark sidebar on the left with the title 'Admin' and a list of links: 'Add Doctor Profile', 'View Doctor Profile', 'View Patient', and 'Electronic Health Records'. The main content area has a header with a hamburger menu icon, the text 'Home', and a red 'LOGOUT' button. Below the header, the title 'Patient List' is centered. A table with 5 columns and 9 rows follows. The columns are labeled '#', 'First Name', 'Last Name', 'Email', and 'Age'. The rows contain patient data, including names like krishana, ganpat, aryan, bhagat, mihir, rathin, yash, and dharmik, along with their email addresses and ages.

#	First Name	Last Name	Email	Age
1	krishana	bhat	krishana@gmail.com	34
2	ganpat	patel	ganpat@gmail.com	56
3	aryan	bhat	aryan@gmail.com	45
4	bhagat	patel	bhagat@gmail.com	55
5	mihir	patel	mihir@gmail.com	34
6	rathin	patel	rathin@yopmail.com	23
7	yash	savant	yash@gmail.com	19
8	dharmik	patel	dharmik@gmail.com	19

Reset password

The screenshot shows a web browser window with the address bar displaying 'localhost:5173/forgetpassword'. The main content area features a centered white card with rounded corners and a subtle drop shadow. The card contains the following elements:

- Title:** 'Forgot Password' in a bold, black font.
- Text:** 'We'll send a password reset link to your email' in a smaller, regular black font.
- Section Header:** 'Email' in a bold, black font.
- Input Field:** A text input box with the placeholder text 'Enter your email'.
- Button:** A black button with the text 'Send Reset Link' in white.
- Text:** 'Remembered your password? [Go to Login](#)' in a regular black font.

The screenshot shows a web browser window with the address bar displaying a URL for a password reset page. The page content is centered and features a 'Reset Password' form. The form includes a title, a subtitle, two input fields for password entry, a submit button, and a link to return to the login page.

Reset Password

Enter your new password below

New Password

Enter new password

Confirm Password

Confirm new password

Reset Password

[Back to Login](#)

The Forgot Password page lets users request a password reset link by entering their registered email. The Reset Password page allows users to securely set a new password using the link sent to their email.

Conclusion

Docon simplifies and modernizes healthcare delivery

By integrating appointments, EHR, telemedicine, and e-prescriptions into one secure platform, Docon enhances efficiency for doctors and accessibility for patients.

A step toward smarter, data-driven healthcare

digital workflows, and seamless integrations, Docon empowers healthcare providers to deliver better, faster, and more personalized care.

Scalable, secure, and future-ready

Built on robust cloud architecture with a focus on privacy and interoperability, Docon is designed to grow with evolving healthcare needs.

Future Enhancement of Implementation

- Enhanced healthcare accessibility through telemedicine.
- Efficient patient management with secure digital health records.
- Reduced waiting times and improved doctor-patient interactions.
- Open APIs for lab, pharmacy, insurance, and IoT health devices.
- AI-driven suggestions for follow-ups & treatment insights.

References

- **MongoDB:**

MongoDB, Inc. "MongoDB Documentation." Available at: <https://docs.mongodb.com/> MongoDB is the NoSQL database used for storing doctor profile data, user information, and patient history in the project.

- **Express.js:**

Express.js Team. "Express Documentation." Available at: <https://expressjs.com/> Express is used for creating the backend API, handling HTTP requests, routes, and middleware for the hospital management system.

- **Node.js:**

Node.js Foundation. "Node.js Documentation." Available at: <https://nodejs.org/en/docs/> Node.js is the runtime environment used for the backend server to execute JavaScript code.

- **React.js:**

- React Team. "React Documentation." Available at: <https://reactjs.org/docs/getting-started.html> React is used for building the user interface of the Docon.